TASK 1: FORMULATING THE WORLD AS AN MDP

Scenario: Smart Traffic Signal Control

A smart traffic signal system controls a busy road intersection. The objective is to reduce traffic congestion while maintaining smooth traffic flow and pedestrian safety.

1. State Space (S)

State	Meaning
Light	Low traffic
Moderate	Normal traffic flow
Heavy	High traffic congestion

S = {Light, Moderate, Heavy}

2. Action Space (A)

Action	Meaning
KeepGreen	Continue current green signal
SwitchSignal	Change traffic direction
PedestrianMode	Enable pedestrian crossing

A = {KeepGreen, SwitchSignal, PedestrianMode}

3. Reward Function (R)

Situation	Reward
Smooth traffic flow	+10
Reduced congestion	+6
Pedestrian crossing success	+4
Heavy congestion	-8

4. Manual Value Iteration

Discount factor: $\gamma = 0.9$

Initial values: V■(s)=0

Two iterations of value iteration are performed.

Final Value Table

State	V■	V■
Light	9.2	17.264
Moderate	8	15.74
Heavy	3.2	9.536

Conclusion

The Smart Traffic Signal Control problem was modeled as a Markov Decision Process (MDP). Using value iteration, the optimal values improved after each iteration, demonstrating how reinforcement learning techniques can optimize traffic management systems.